HeatAtlas: Forecast Heat. Prioritise Relief.

Autonomous Extreme Heat Relief Engine & Hyper-Local Spatial Triage for the Global South

HACKATHON TRACK

Theme 2: Climate, Resource Infrastructure & Global Supply Resi
lience

PROBLEM SPECTRUM

Autonomous Emergency Response & Supply Chain Logistics Swarm + Smart Grid & Urban Resource Allocation

CORE ENGINE

Deterministic Python Spatial Math (Eh, Vh, Rd) + Google Gemini 2.5 Flash Autonomous Tool-Calling Planner

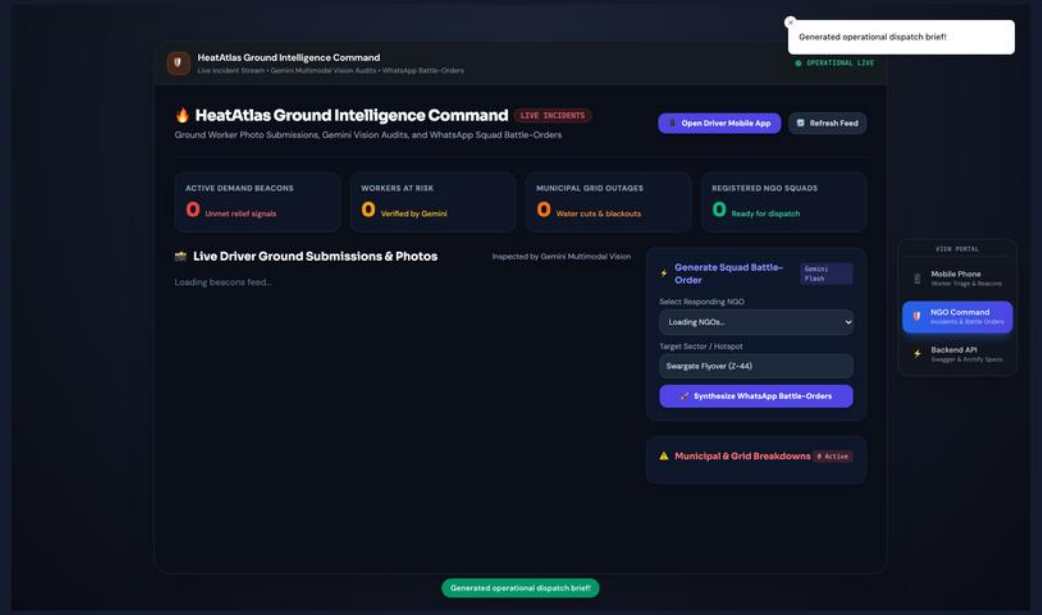
LIVE DEPLOYMENT

Production Web App: https://heatatlas.vercel.app | GitHub: github.com/vishwjeetlol/heatatlas

64
Micro-Zones (Pune)

100%
Live APIs (Zero Mock)

< 60s
Dispatch Brief Time



Live Production App: Mobile Citizen Map + NGO Command Center + FastAPI Backend

Why City-Wide Weather Alerts Fail During 44°C Heatwaves

When Pune, Nagpur, or Delhi hits 44°C, heat stress and water scarcity strike hyper-locally by neighborhood.

1. The Single-Number Weather Blindspot

Standard weather apps report one city-wide temperature ('40°C Pune'). A delivery rider on Swargate asphalt faces 44.2°C radiant heat with broken taps, while shaded neighborhoods have working relief.

2. 10M+ Outdoor Workers Exposed

Delivery riders, street vendors, traffic police, and construction workers spend 8–10 hours in unshaded transit the corridors without knowing nearest verified working drinking water tap or ORS clinic.

3. Cascading Water & Grid Infrastructure Failures

Extreme afternoon heat triggers MSEDCL transformer outages and commercial RO chiller failures—shutting off water and cooling simultaneously when thermal vulnerability peaks.

4. Blind NGO Relief Dispatch & Fatal Triage Delays

Volunteer leads allocate 20L water cans and ORS packets by guesswork instead of mathematical deficit. Meanwhile, victims miss early anhidrosis (stopped sweating) signs before fatal heatstroke.

End-to-End Spatial Intelligence & Autonomous Relief Dispatch

Two tailored portals—Outdoor Worker Mobile Triage & NGO Volunteer Lead Command—backed by live APIs.

64-Zone Relief Deficit (Rd) Map

Splits Pune (18.41,73.70 to 18.65,73.95) into an 8x8 grid to (Z-00 Z-77). Ranks every zone by hourly Heat Exposure (Eh), Demographic Vulnerability (Vh), and Relief Deficit (Rd).

Multilingual AI Dispatch Planner

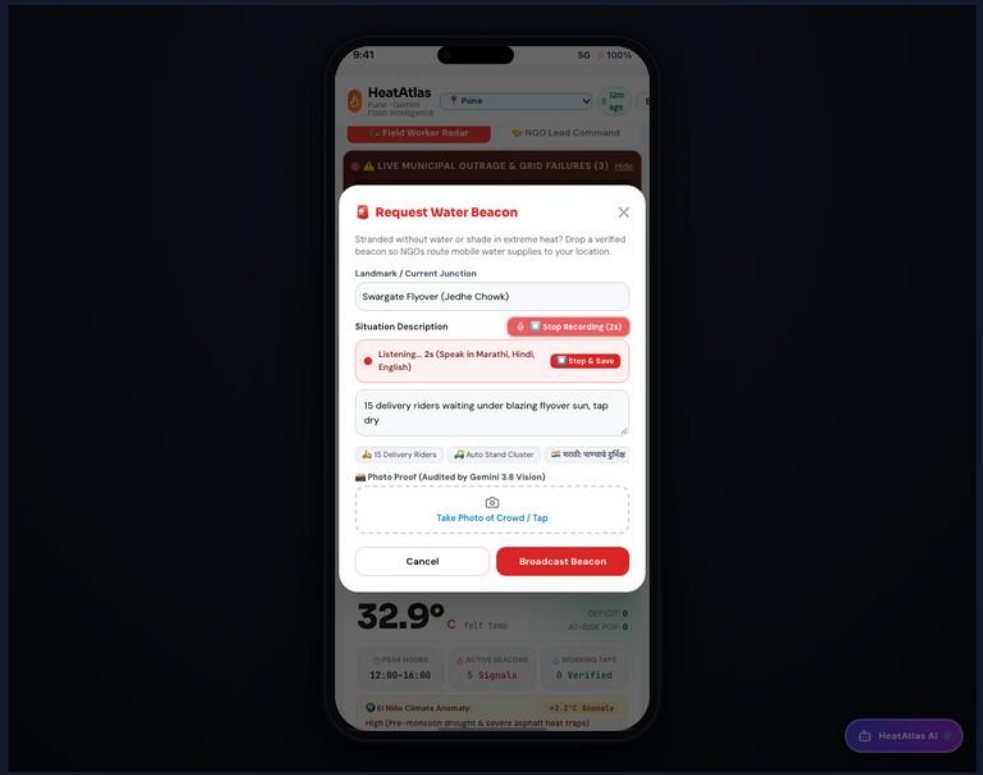
Volunteer leads speak or type in English, Hindi, or Marathi 2 ('Send teams to Yerwada, 40 water cans'). Gemini 2.5 Flash and calls live zone/asset tools outputs a structured WhatsApp brief.

Strict Gemini Vision Anti-Spoofing

Workers upload photos of dry/broken taps or crowd heat Gemini beacons. Vision audits image authenticity first—rejecting real random/spoof photos while verifying water infrastructure.

3-Tap Relief Finder & 108 Triage

Workers tap 'Nearest Relief' for the 3 closest working water & taps clinics via Haversine distance, or use the 2-tap symptom Anhidrosis heatstroke checker routing directly to Call 108.



Worker View: 64-Zone Map, Nearest Verified Taps & Voice Intake

Zero-Hallucination Split: Deterministic Python + Agentic Gemini

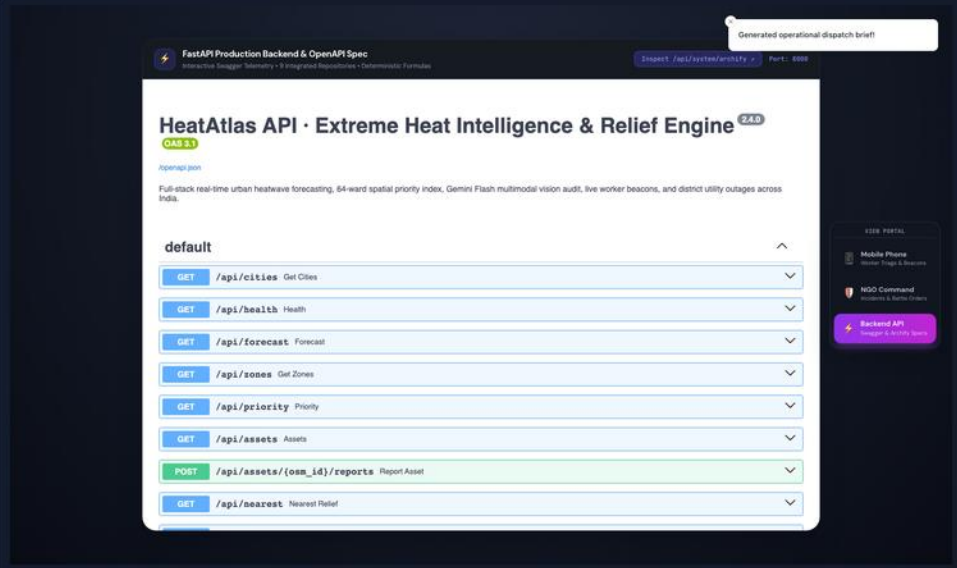
Python computes 100% of heat/vulnerability math; Gemini handles NL extraction, tool routing & brief synthesis.

1. DETERMINISTIC SCORING ENGINE (services/scoring.py)

- Heat Exposure: Eh = Pop × hours(T_apparent >= 38°C) over next 24h
- Vulnerability: Vh = Eh × (1 + (ChildPop_0_5 + ElderlyPop_65+) / Pop)
- Relief Deficit: Rd = Vh / (1 + WorkingTaps + 2 × Clinics)
- Zero LLM involvement in numerical computation — impossible to hallucinate metrics.

2. AGENTIC TOOL-CALLING & GUARDRAILS (services/agent.py)

- Step 1 Intake: Gemini 2.5 Flash extracts {intent, zone_hint, teams, water_cans_l, ors, window}
- Step 2 Tool Loop: Calls get_top_deficit_zones, get_zone_detail, get_nearest_relief
- Step 3 Hallucination Guard: validate_dispatch_brief() verifies every zone_id & Rd against Python sta
- Step 4 Anti-Spoof Vision: Image-first verification rejects non-tap/non-crowd uploads automatically.



Live FastAPI OpenAPI Backend (/api/zones, /api/dispatch, /api/report)

100% Real-Time Open Datasets & Live Persistence

Built under FORGE'26's Zero-Tolerance Dummy Data Policy — every zone and asset comes from live upstream APIs.

Open-Meteo Forecast API (Live, 30m TTL)

Fetches hourly temperature_2m, relative_humidity_2m, and apparent_temperature for all 64 Pune zone centroids a
cross a 72-hour window.

WorldPop v2 Demographic API (24h Cache)

Retrieves real population, under-5 child population, and over-65 elderly population rasters per zone bounding
box with automatic retry.

OpenStreetMap Overpass API (6h Cache)

Queries live amenity=drinking_water, amenity=water_point, amenity=clinic, and amenity=hospital nodes across Pu
ne (18.41,73.70 to 18.65,73.95).

SQLite Live Crowdsourced State (heatatlas.db)

Persists citizen broken-tap reports (with 30-min duplicate rate-limiting HTTP 429) and recalculates zone Relie
f Deficit (Rd) in < 2 seconds.

Ready for Immediate Deployment Across Heat-Vulnerable Cities

Tested across 9 automated acceptance criteria suites and deployed live on Vercel.

Phase 1 (Live Today — Pune 64-Zone Grid)

Full-stack FastAPI + React + Leaflet app live at https://heatatlas.vercel.app. 9/9 pytest acceptance criteria pa
ssing (AC-1 through AC-6 + anti-spoof vision).

Phase 2 (Multi-City Heat Corridor Expansion)

Pre-configured city bounding boxes ready for Nagpur, Ahmedabad, Delhi NCR, and Chennai with IMD Heat Action Plan
(HAP) threshold calibration.

Phase 3 (Municipal & ONDC Logistics Integration)

Direct webhook dispatch to municipal water tanker fleets (PMC/MSEDCL) and automated ORS/hydration restocking via
local pharmacy networks.

Live Links for Judges

Web App: https://heatatlas.vercel.app | GitHub: https://github.com/vishwjeetlol/heatatlas | Demo Video: http
s://heatatlas.vercel.app/demo.mp4